

VARSHA JOHN

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ABOUT ME

Computer Science & Engineering graduate with strong Python fundamentals and extensive hands-on experience in full-stack MERN development. Proven capability in architecting end-to-end software solutions, ranging from AI-driven mobile health applications to secure, decentralized blockchain networks. Equipped with a deep understanding of core CS fundamentals, RESTful API design, and database management. A proactive problem-solver and collaborative team lead ready to contribute to scalable engineering teams.

TECHNICAL SKILLS

- Languages: **Python, C, SQL, HTML5, CSS3**
- Frameworks & Libraries: **React.js, Node.js, Express.js, React Native, Flask**
- Databases & Tools: **MongoDB, SQLite, Git, GitHub, RESTful APIs**
- AI & Advanced Tech: **YOLO v8 (Object Detection), Groq API (LLAMA), Blockchain Fundamentals**
- Core CS Concepts: **Data Structures & Algorithms (DSA), Database Management Systems (DBMS), Object-Oriented Programming (OOPs)**

EDUCATION

2022 - 2026	B.Tech in Computer Science & Engineering
Marian Engineering College, Trivandrum	CGPA:8.65
2022	Higher Secondary (Class XII)
Pallithura Higher Secondary School, Trivandrum	96.33%
2020	Secondary (Class X)
Pallithura Higher Secondary School, Trivandrum	90%

PROJECTS

NUTRISNAP: AI-DRIVEN NUTRITION ASSISTANT

A full-stack AI-powered mobile health application designed to analyze food, track nutrition, and provide personalized dietary insights. The system integrates image recognition, barcode scanning, and AI-driven recommendations to deliver real-time health analysis and smart nutrition guidance.

Contribution:

- Engineered the computer vision pipeline by training a custom YOLO model for food item detection and integrating it into the Express.js API.
- Built a dynamic recommendation engine using the Groq API (LLaMA) to parse user goals and medical conditions into structured dietary plans.
- Built health risk analysis features to assess conditions like obesity, diabetes, and hypertension using dietary patterns.
- Added key UI features such as smart risk visualization and streak-based daily tracking for user engagement.

Tech Stack: React Native, Node.js, Express.js, MongoDB, Python (YOLO), Groq API (LLaMA), Expo, GitHub

ProjectLink: <https://github.com/varshajohn/NutriSnap-Final.git>

BLOCKCHAIN-BASED E-VOTING SYSTEM

A secure and transparent e-voting system leveraging blockchain technology to ensure immutability and prevent vote tampering. The application supports voter registration, nominee management, secure authentication, vote casting, and real-time result display with an easy-to-use web interface.

Contribution:

- Built a custom backend ledger implementing SHA-256 cryptographic hashing to secure voting workflows and guarantee data immutability.
- Programmed election verification logic within Flask controller routes to prevent duplicate voting and assure data integrity across nodes.
- Created an asynchronous results dashboard using JavaScript and HTML/CSS to query an SQLite database for live voter analytics.

Tech Stack: Python, Flask, SQLite, Custom Blockchain (SHA-256), HTML, CSS, JavaScript

ProjectLink: https://github.com/varshajohn/Blockchain_Voting.git

PLANWISE – A TASK TRACKER APP

A full-stack task tracking application built to streamline project and team management. It supports secure admin and team member logins, task assignments, status tracking, and dashboards for real-time project monitoring. Features include project creation, team management, task allocation, and role-based access for efficient collaboration.

Contribution:

- Developed an Admin Panel with project, team, and task management modules, including authentication and password management for both admins and team members.
- Built responsive frontend dashboards using React, MUI, and Tailwind, and implemented backend REST APIs for authentication, task handling, and project management.

Tech Stack: MERN (MongoDB, Express.js, React.js, Node.js), MUI, Tailwind, GitHub

ProjectLink: <https://github.com/varshajohn/PLANWISE.git>

CERTIFICATIONS

DBMS(NPTEL, 8 weeks) – Covered SQL, ER modeling, normalization, and transaction management.